

BPEL Code:

```
<process>
  <sequence>
    <receive name="New patient" createInstance="yes"/>
    <scope>
      <!-- Variables -->
      <variables>
        <variable name="isApproved" type="xsd:boolean"/>
      </variables>
      <!-- Initial Value -->
      <assign>
        <copy>
          <from>false()</from>
          <to>$isApproved</to>
        </copy>
      </assign>

      <while condition="$isApproved = false()">
        <sequence>
          <invoke name="Request an appointment"/>

          <!-- Mapping tc1 -->
          <switch>
            <case condition="Normal">
              <invoke name="Set with earlier Health provider"/>
            </case>
            <case condition="Emergency">
              <invoke name="Set with closest Health provider"/>
            </case>
          </switch>

          <!-- Mapping tc2 -->
          <flow>
            <switch>
              <case condition="Email">
                <invoke name="Notify by email"/>
              </case>
              <case condition="SMS">
                <invoke name="Notify by SMS"/>
              </case>
              <case condition="Email and SMS">
                <invoke name="Notify by email"/>
                <invoke name="Notify by SMS"/>
              </case>
            </switch>
          </flow>

          <switch>
            <case condition="Cancel appointment">
              <invoke name="Cancel appointment"/>
              <switch>
                <case condition="Exit">
                  <reply name="Patient left"/>
                  <terminate/>
                </case>
              </switch>
            </case>

            <case condition="Approve appointment">
              <assign>
                <copy>
                  <from>true()</from>
                  <to>$isApproved</to>
                </copy>
              </assign>
            </case>
          </switch>
        </sequence>
      </while>
    </scope>
  </sequence>
</process>
```

```
        <case condition="Change appointment">
            <invoke name="Change appointment"/>
        </case>
    </switch>
</sequence>
</while>
</scope>

<!-- Mapping tc3 -->
<invoke name="Send information to Health provider"/>
<invoke name="Track the care of the patient"/>
<invoke name="Check for insurance"/>

<!-- Mapping tc6 -->
<switch>
    <case condition="Yes">
        <invoke name="Pay by insurance company"/>
    </case>
    <case condition="No">
        <invoke name="Pay by credit card"/>
    </case>
</switch>
<reply name="Patient served"/>
</sequence>
</process>
```